```
/*Name:Yuvraj Vinayak Suryawanshi
Roll no:181
Div:IT C*/
#include<iostream>
#include<string.h>
using namespace std;
int n;
int roll[20];
string name[20];
void linersearch(int key)
{
for(int i=0;i<n;i++)
{
if(roll[i]==key)
{
cout<<"key is found";
cout<<"the roll no is"<<roll[i]<<endl;
cout<<"the name is"<<name[i]<<endl;return;
}
}
cout<<"key is not found";
}
void binarysearch(int key){
int low=0;
int high=n-1;
while(low<=high){
int mid=(low+high)/2;
if(roll[mid]==key)
{
cout<<"student is found"<<endl;
cout<<"roll no is:"<<roll[mid]<<endl;
cout<<"name is:"<<name[mid]<<endl;return;
}
else if(key<roll[mid])
high=mid-1;
else
low=mid+1;
}
cout<<"student is not found";
}
int main(){
int choice ;
int key;
cout<<" enter number of student:";
cin>>n;
for(int i=0;i<n;i++)
{
cout<<"enterthe roll number:"<<endl;cin>>roll[i];
```

```
cout<<"enterthe string:"<<endl;
cin>>name[i];
}
do{
cout<<"\n\n-------menu-----";
cout<<"\n1.linear search";
cout<<"\n2.binary search";
cout<<"\n3.exit"<<endl;
cout<<"enter your choice"<<endl;
cin>>choice;
if(choice==1)
{
cout<<"enter your key:"<<endl;cin>>key;
linersearch(key);
}
else if(choice==2)
{
cout<<"enter your key:"<<endl;cin>>key;
binarysearch(key);
}
else if(choice==3)
{
cout<<"THANK YOU"<<endl;
}
else{
cout<<"choice is not found:";
}
}
while( choice!=3);
return 0;
}
/* pl@pl-OptiPlex-3020:~$ g++ Yuvraj_DSA1.cpp
pl@pl-OptiPlex-3020:~$ ./a.out enter number of student:2
enterthe roll number:
100
enterthe string:
athrva
enterthe roll number:
79
enterthe string:
Yuvraj
-------menu-----
1.linear search
2.binary search
3.exit
enter your choice1 enter your key:
100
student is found
```

```
roll no is:100
name is:athrva
-------menu-----
1.linear search
2.binary search
3.exit
enter your choice2 enter your key:
100
student is found
roll no is:100
name is:athrva
-------menu-----
1.linear search
2.binary search
3.exit
enter your choice4 choice is not found:
-------menu-----
1.linear search
2.binary search
3.exit
enter your choice3
THANK YOU
*/
```